

Lem cuerpo imp di

Lema 1. *Todo cuerpo es un dominio de integridad.*

Demostración: Sea $(K, +, \cdot)$ un cuerpo y $a, b \in K$ tales que $a \cdot b = 0$. Por contradicción, supongamos que $a \neq 0 \wedge b \neq 0$. Como $a \neq 0$ y K es un cuerpo, $\exists a^{-1} \in K : a \cdot a^{-1} = 1$.

$$a \cdot b = 0 \implies a^{-1} \cdot a \cdot b = a^{-1} \cdot 0 \implies 1 \cdot b = 0 \implies b = 0 \quad \longrightarrow \longleftarrow$$



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